Quadratics & Nonlinear Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the community theater arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?

Answer: -3. From the labeled graph, the vertex of the parabola is $(1, -4)$ and the parabola passes through $(2, -3)$. Using vertex form $q(x) = a(x - 1)^2 + (-4)$, substituting $(2, -3)$ gives $-3 = a(1)^2 + (-4) \Rightarrow a = 1$. Evaluating at $x = 0$ gives $q(0) = 1(0 - 1)^2 + (-4) = -3$.

2 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the mountain observatory arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?

Answer: 1. From the labeled graph, the vertex of the parabola is $(-1, -1)$ and the parabola passes through $(0, 1)$. Using vertex form $q(x) = a(x - -1)^2 + (-1)$, substituting $(0, 1)$ gives $1 = a(1)^2 + (-1) \Rightarrow a = 2$. Evaluating at $x = 0$ gives $q(0) = 2(0 - -1)^2 + (-1) = 1$.

3 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the solar microgrid facility arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?

Answer: -1. From the labeled graph, the vertex of the parabola is $(2, -5)$ and the parabola passes through $(3, -4)$. Using vertex form $q(x) = a(x - 2)^2 + (-5)$, substituting $(3, -4)$ gives $-4 = a(1)^2 + (-5) \Rightarrow a = 1$. Evaluating at $x = 0$ gives $q(0) = 1(0 - 2)^2 + (-5) = -1$.

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4 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the publishing house arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?

Answer: 6. From the labeled graph, the vertex of the parabola is $(2, -2)$ and the parabola passes through $(3, 0)$. Using vertex form $q(x) = a(x-2)^2 + (-2)$, substituting $(3, 0)$ gives $0 = a(1)^2 + (-2) \Rightarrow a = 2$. Evaluating at $x = 0$ gives $q(0) = 2(0-2)^2 + (-2) = 6$.

5 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the hydroponic greenhouse arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?

Answer: 3. From the labeled graph, the vertex of the parabola is $(3, -6)$ and the parabola passes through $(4, -5)$. Using vertex form $q(x) = a(x-3)^2 + (-6)$, substituting $(4, -5)$ gives $-5 = a(1)^2 + (-6) \Rightarrow a = 1$. Evaluating at $x = 0$ gives $q(0) = 1(0-3)^2 + (-6) = 3$.

6 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the robotics club arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?

Answer: -1. From the labeled graph, the vertex of the parabola is $(1, -3)$ and the parabola passes through $(2, -1)$. Using vertex form $q(x) = a(x-1)^2 + (-3)$, substituting $(2, -1)$ gives $-1 = a(1)^2 + (-3) \Rightarrow a = 2$. Evaluating at $x = 0$ gives $q(0) = 2(0-1)^2 + (-3) = -1$.

7 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the wildlife sanctuary arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?

Answer: 1. From the labeled graph, the vertex of the parabola is $(-1, 0)$ and the parabola passes through $(0, 1)$. Using vertex form $q(x) = a(x--1)^2 + (0)$, substituting $(0, 1)$ gives $1 = a(1)^2 + (0) \Rightarrow a = 1$. Evaluating at $x = 0$ gives $q(0) = 1(0--1)^2 + (0) = 1$.

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

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8 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the robotics club, the quadratic function $h(u) = -u^2 + 28u - 140$ attains its maximum at $u = k$. What is the maximum value $h(k)$?

Answer: 56. The vertex occurs at $k = 14$, giving $h(14) = 196 - 140 = 56$.

9 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the marine research vessel, in a calculation for the local farmers market (case 20), if $f(x) = x^2 - 6x - 2$, what is $f(5)$?

Answer: -7. Applying Equivalent Expressions & Quadratic Equations principles, $f(5) = 1(5)^2 - 6(5) - 2 = -7$ Therefore, the verified correct option is -7.

10 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the campus coffee roastery, for all $x \neq 5$, the rational expression $\frac{x^2 - 25}{x - 5}$ simplifies to $x + f$. What is the scaled constant $3f$?

Answer: 15. Factoring gives $x + 5$, so $f = 5$ and $3f = 15$.

11 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the meteorological station, in a calculation for the coastal aquarium (case 28), if $f(x) = 2x^2 + 3x + 5$, what is $f(1)$?

Answer: 10. Applying Equivalent Expressions & Quadratic Equations principles, $f(1) = 2(1)^2 + 3(1) + 5 = 10$ Therefore, the verified correct option is 10.

12 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the urban bike-share station, in a calculation for the public transit depot (case 32), if $f(x) = 5x^2 + 3$, what is $f(4)$?

Answer: 83. Applying Equivalent Expressions & Quadratic Equations principles, $f(4) = 5(4)^2(4) + 3 = 83$ Therefore, the verified correct option is 83.

13 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the art gallery, in a calculation for the wildlife sanctuary (case 36), if $f(x) = x^2 - 8x + 6$, what is $f(5)$?

Answer: -9. Applying Equivalent Expressions & Quadratic Equations principles, $f(5) = 1(5)^2 - 8(5) + 6 = -9$ Therefore, the verified correct option is -9.

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14 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the wind energy cooperative, in a calculation for the community orchard (case 44), if $f(x) = 3x^2 - x + 7$, what is $f(-5)$?

Answer: 87. Applying Equivalent Expressions & Quadratic Equations principles, $f(-5) = 3(-5)^2 - 1(-5) + 7 = 87$ Therefore, the verified correct option is 87.

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the regional airport shuttle, in a calculation for the historic lighthouse (case 68), what is the product of the solutions to $2x^2 + 6x - 20 = 0$?

Answer: -10. By Vieta's formulas, product of roots $= \frac{c}{a} = \frac{-20}{2} = -10$.

16 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

For the botanical conservatory parabolic reflector equation $1x^2 - bx + 9 = 0$, b is a positive constant. If the equation has exactly one real solution, what is the value of b?

Answer: 6. A quadratic equation $ax^2 - bx + c = 0$ has exactly one real solution when its discriminant equals zero: $b^2 - 4(1)(9) = 0 \Rightarrow b^2 = 36$. Since $b > 0$, $b = 6$.

17 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $x + y = 9$ and $x^2 - y^2 = 27$ in a calibration formula at the municipal library, what is the value of $x - y$?

Answer: 3. Factoring the difference of squares gives $x^2 - y^2 = (x + y)(x - y)$. Substituting $x + y = 9$ yields $9(x - y) = 27 \Rightarrow x - y = \frac{27}{9} = 3$.

18 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the product of the solutions to the quadratic equation $4x^2 - 7x + 20 = 0$ used in the community theater design model?

Answer: 5. For any quadratic equation $ax^2 + bx + c = 0$, the product of the solutions is $\frac{c}{a} = \frac{20}{4} = 5$.

19 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

For the regional bakery parabolic reflector equation $1x^2 - bx + 36 = 0$, b is a positive constant. If the equation has exactly one real solution, what is the value of b?

Answer: 12. A quadratic equation $ax^2 - bx + c = 0$ has exactly one real solution when its discriminant equals zero: $b^2 - 4(1)(36) = 0 \Rightarrow b^2 = 144$. Since $b > 0$, $b = 12$.

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20 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $x + y = 12$ and $x^2 - y^2 = 72$ in a calibration formula at the coastal aquarium, what is the value of $x - y$?

Answer: 6. Factoring the difference of squares gives $x^2 - y^2 = (x + y)(x - y)$. Substituting $x + y = 12$ yields $12(x - y) = 72 \Rightarrow x - y = \frac{72}{12} = 6$.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

21 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the botanical conservatory arch design is shown in the xy -plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?

a. -5 ✓

b. -3

c. -6

d. -4

From the labeled graph, the vertex of the parabola is $(-1, -6)$ and the parabola passes through $(0, -5)$. Using vertex form $q(x) = a(x - -1)^2 + (-6)$, substituting $(0, -5)$ gives $-5 = a(1)^2 + (-6) \Rightarrow a = 1$. Evaluating at $x = 0$ gives $q(0) = 1(0 - -1)^2 + (-6) = -5$.

22 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the municipal library arch design is shown in the xy -plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?

a. 5

b. -5

c. -1

d. 3 ✓

From the labeled graph, the vertex of the parabola is $(2, -5)$ and the parabola passes through $(3, -3)$. Using vertex form $q(x) = a(x - 2)^2 + (-5)$, substituting $(3, -3)$ gives $-3 = a(1)^2 + (-5) \Rightarrow a = 2$. Evaluating at $x = 0$ gives $q(0) = 2(0 - 2)^2 + (-5) = 3$.

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23 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the regional bakery arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?

a. 1

b. 5 ✓

c. 7

d. -3

From the labeled graph, the vertex of the parabola is $(2, -3)$ and the parabola passes through $(3, -1)$. Using vertex form $q(x) = a(x-2)^2 + (-3)$, substituting $(3, -1)$ gives $-1 = a(1)^2 + (-3) \Rightarrow a = 2$. Evaluating at $x = 0$ gives $q(0) = 2(0-2)^2 + (-3) = 5$.

24 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the coastal aquarium arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?

a. 7 ✓

b. 9

c. -2

d. 2

From the labeled graph, the vertex of the parabola is $(3, -2)$ and the parabola passes through $(4, -1)$. Using vertex form $q(x) = a(x-3)^2 + (-2)$, substituting $(4, -1)$ gives $-1 = a(1)^2 + (-2) \Rightarrow a = 1$. Evaluating at $x = 0$ gives $q(0) = 1(0-3)^2 + (-2) = 7$.

25 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the science museum arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?

a. 0

b. 3

c. 4 ✓

d. 6

From the labeled graph, the vertex of the parabola is $(2, 0)$ and the parabola passes through $(3, 1)$. Using vertex form $q(x) = a(x-2)^2 + (0)$, substituting $(3, 1)$ gives $1 = a(1)^2 + (0) \Rightarrow a = 1$. Evaluating at $x = 0$ gives $q(0) = 1(0-2)^2 + (0) = 4$.

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26 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the urban bike-share station arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?

a. -3

b. -4 ✓

c. -2

d. -6

From the labeled graph, the vertex of the parabola is $(1, -6)$ and the parabola passes through $(2, -4)$. Using vertex form $q(x) = a(x - 1)^2 + (-6)$, substituting $(2, -4)$ gives $-4 = a(1)^2 + (-6) \Rightarrow a = 2$. Evaluating at $x = 0$ gives $q(0) = 2(0 - 1)^2 + (-6) = -4$.

27 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the ceramic studio arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?

a. 16

b. -4

c. 1

d. 14 ✓

From the labeled graph, the vertex of the parabola is $(3, -4)$ and the parabola passes through $(4, -2)$. Using vertex form $q(x) = a(x - 3)^2 + (-4)$, substituting $(4, -2)$ gives $-2 = a(1)^2 + (-4) \Rightarrow a = 2$. Evaluating at $x = 0$ gives $q(0) = 2(0 - 3)^2 + (-4) = 14$.

28 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph of the quadratic function $y = q(x)$ for the agricultural cooperative arch design is shown in the xy-plane, where the vertex and a second point on the parabola are labeled. What is the y-intercept $q(0)$ of the parabola?

a. -3

b. -1

c. -2 ✓

d. 0

From the labeled graph, the vertex of the parabola is $(-1, -3)$ and the parabola passes through $(0, -2)$. Using vertex form $q(x) = a(x - -1)^2 + (-3)$, substituting $(0, -2)$ gives $-2 = a(1)^2 + (-3) \Rightarrow a = 1$. Evaluating at $x = 0$ gives $q(0) = 1(0 - -1)^2 + (-3) = -2$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

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29 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the art gallery, consider the cubic polynomial $f(x) = x^3 - 5x^2 + 10x - 20$. What is the remainder $f(2)$ upon division by $(x - 2)$?

a. -22

b. -12 ✓

c. -7

d. -10

By the Remainder Theorem, $f(2) = 2^3 - 5(2^2) + 10(2) - 20 = -12$.

30 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the culinary institute, the exponential equation $6^{2x-1} = 6^{4+5}$ holds for real x. What is the quantity $f = 2x$?

a. 14

b. 5

c. 10 ✓

d. 16

Equating exponents gives $2x - 1 = 9 \Rightarrow 2x = 10$.

31 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the water treatment plant, in a calculation for the textile workshop (case 14), what is the vertex of the parabola $y = x^2 - 14x + 53$?

a. (7, -4)

b. (-7, 4)

c. **(7, 4)** ✓

d. (8, 4)

Vertex $x = -\frac{b}{2a} = 7$, $y = f(7) = 4$. Vertex $= (7, 4)$.

32 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the sports complex, the radical function $f(x) = \sqrt{7x + 11}$ is defined for $7x + 11 \geq 0$. What is the product $7x$ when $f(x) = 9$?

a. 80

b. 59

c. 70 ✓

d. 77

Squaring both sides yields $7x + 11 = 81 \Rightarrow 7x = 70$.

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33 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the maple syrup farm, in a calculation for the honeybee apiary (case 16), how many x-intercepts does the graph of $y = -2x^2 - 5x - 3$ have?

a. Zero

b. One

c. **Two ✓**

d. Cannot be determined

Discriminant $= b^2 - 4ac = (-5)^2 - 4(-2)(-3) = 1$, which is positive, giving two real intercept(s).

34 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the planetarium, in a calculation for the carpentry workshop (case 17), how many x-intercepts does the graph of $y = -x^2 - x + 2$ have?

a. Zero

b. One

c. **Two ✓**

d. Cannot be determined

Discriminant $= b^2 - 4ac = (-1)^2 - 4(-1)(2) = 9$, which is positive, giving two real intercept(s).

35 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the regional airport shuttle, in a calculation for the historic lighthouse (case 18), how many x-intercepts does the graph of $y = x^2 - 2x + 4$ have?

a. **Zero ✓**

b. One

c. Two

d. Cannot be determined

Discriminant $= b^2 - 4ac = (-2)^2 - 4(1)(4) = -12$, which is negative, giving zero real intercept(s).

36 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the wind energy cooperative, in a calculation for the community orchard (case 19), if $f(x) = 2x^2 + 2x - 2$, what is $f(5)$?

a. 63

b. 58 ✓

c. 49

d. 66

Applying Equivalent Expressions & Quadratic Equations principles, $f(5) = 2(5)^2 + 2(5) - 2 = 58$

Therefore, the verified correct option is 58.

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37 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the national park visitor center, in a calculation for the environmental club (case 21), if $f(x) = 4x^2 + 2x + 7$, what is $f(-4)$?

a. 63 ✓

b. 66

c. 57

d. 62

Applying Equivalent Expressions & Quadratic Equations principles, $f(-4) = 4(-4)^2 + 2(-4) + 7 = 63$ Therefore, the verified correct option is 63.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the maple syrup farm, for what sum of x-values does the graph of $y = x^2 + x - 12$ cross the x-axis?

a. -6

b. -1 ✓

c. 3

d. 2

The x-intercepts are $x = -4$ and $x = 3$ (from factoring), so their sum is -1.

39 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the planetarium, in a calculation for the carpentry workshop (case 67), what is the sum of the solutions to $2x^2 - 4x - 70 = 0$?

a. 1

b. -7

c. 2 ✓

d. -3

By Vieta's formulas, sum of roots $= -\frac{b}{a} = -\frac{-4}{2} = 2$.

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40 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the wind energy cooperative, the quadratic graph $p(t) = 6t^2 + qt + 6$ is tangent to the horizontal axis with $q > 0$. What is the positive coefficient q?

a. 13

b. 18

c. 12 ✓

d. 3

Setting discriminant $\Delta = q^2 - 4(6)(6) = 0$ gives $q = 12$.

41 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In a model for the marine research vessel, for what sum of x-values does the graph of $y = -x^2 + 4x + 12$ cross the x-axis?

a. 7

b. 5

c. 11

d. 4 ✓

The x-intercepts are $x = 6$ and $x = -2$ (from factoring), so their sum is 4.

42 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

For the university lecture hall parabolic reflector equation $1x^2 - bx + 49 = 0$, b is a positive constant. If the equation has exactly one real solution, what is the value of b?

a. 49

b. 18

c. 14 ✓

d. 7

A quadratic equation $ax^2 - bx + c = 0$ has exactly one real solution when its discriminant equals zero: $b^2 - 4(1)(49) = 0 \Rightarrow b^2 = 196$. Since $b > 0$, $b = 14$.

43 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $x + y = 17$ and $x^2 - y^2 = 102$ in a calibration formula at the culinary institute, what is the value of $x - y$?

a. 85

b. 6 ✓

c. 11

d. 8

Factoring the difference of squares gives $x^2 - y^2 = (x + y)(x - y)$. Substituting $x + y = 17$ yields $17(x - y) = 102 \Rightarrow x - y = \frac{102}{17} = 6$.

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44 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the product of the solutions to the quadratic equation $2x^2 - 25x + 14 = 0$ used in the wildlife sanctuary design model?

a. 7 ✓

b. 14

c. 25

d. 9

For any quadratic equation $ax^2 + bx + c = 0$, the product of the solutions is $\frac{c}{a} = \frac{14}{2} = 7$.

45 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

For the water treatment plant parabolic reflector equation $1x^2 - bx + 9 = 0$, b is a positive constant. If the equation has exactly one real solution, what is the value of b?

a. 3

b. 9

c. 10

d. 6 ✓

A quadratic equation $ax^2 - bx + c = 0$ has exactly one real solution when its discriminant equals zero: $b^2 - 4(1)(9) = 0 \Rightarrow b^2 = 36$. Since $b > 0$, $b = 6$.

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